

# OPH Cosmological Vacuum and Structure Formation: Finite Transfer Gates, Fluctuation Ensembles, and Proto-Object Receipts

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## Abstract

OPH cosmology needs a sharp distinction between a field that looks noisy and a field that deserves to be called a physical vacuum. This paper sets that boundary. A settled simulator state, a repair relaxation, or a plausible visual fluctuation is only a diagnostic unless it comes from OPH source data with a proper transfer construction and refinement check.

The same boundary organizes structure formation. Vacuum-like fields, primordial seeds, and observable sky predictions are different claims. The paper keeps those layers separate so a useful simulation cannot be mistaken for a completed cosmological prediction.

## 1 Scope

This is a technical companion outside the release bundle. It coordinates the vacuum and fluctuation boundary for finite-source cosmology, screen microphysics, the compact theorem stack, and OPH-FPE simulator outputs.

## 2 Shared Finite Quotient Theorem Surface

**Finite quotient ensemble theorem surface.** Fix a finite regulator  $r$ . The physical presentation space is  $\Sigma_r$ , the presentation redundancy groupoid is  $\Gamma_r$ , and the finite physical quotient is

$$Q_r = \Sigma_r / \Gamma_r, \quad \pi_r : \Sigma_r \rightarrow Q_r.$$

The quotient removes nonphysical presentation data: gauge representatives, port relabelings, mesh labels, shard or worker identifiers, queue order, repair schedule identifiers, retry counters, timestamps unless declared semantic, hidden carrier coordinates, and inert ancillary labels. If only settled configurations carry probability, the probability space is the normal-form subset

$$N_r = n_r(Q_r).$$

The map  $n_r$  is a normal-form map, not a probability law. Any promoted physical branch inherits this firewall: it must declare the quotient-intrinsic source law or action before a normal form can be read as a selection or prediction claim.

**Observable algebras and reference states.** In the finite classical case the quotient observable algebra is

$$\mathcal{O}_r = \ell^\infty(Q_r).$$

In the finite quantum case the physical algebra is a declared quotient algebra  $\mathcal{A}_r^{\text{phys}}$  with state

$$\omega_r(A) = \text{Tr}(\rho_r A).$$

When the reference object is obtained from a finite lifted carrier, the load-bearing data are not an abstract groupoid cardinality alone. They are a tracially pointed quotient

$$\left(\mathcal{A}_{r,b}^{\text{phys}}, \tau_{r,b}^0\right), \quad \mathcal{A}_{r,b}^{\text{phys}} = z_{r,b} B(\tilde{\mathcal{H}}_r)^{G_r} z_{r,b},$$

where  $U_r : G_r \rightarrow U(\tilde{\mathcal{H}}_r)$  is the compact gauge action and  $z_{r,b}$  is the central projection for the declared boundary or superselection sector. The reference trace is

$$\tau_{r,b}^0(A) = \frac{\text{Tr}_{\tilde{\mathcal{H}}_r}(A)}{\text{Tr}_{\tilde{\mathcal{H}}_r}(z_{r,b})}.$$

If

$$z_{r,b} \tilde{\mathcal{H}}_r \cong \bigoplus_{\alpha} V_{\alpha} \otimes M_{\alpha},$$

with  $d_{\alpha} = \dim V_{\alpha}$  and  $m_{\alpha} = \dim M_{\alpha}$ , then

$$\mathcal{A}_{r,b}^{\text{phys}} \cong \bigoplus_{\alpha} I_{V_{\alpha}} \otimes B(M_{\alpha}), \quad p_{r,\alpha} = \frac{d_{\alpha} m_{\alpha}}{\sum_{\beta} d_{\beta} m_{\beta}}.$$

These are the induced central-sector weights only after the carrier representation and boundary sector have been fixed.

**OPH quotient ensemble.** An OPH quotient ensemble is specified by a quotient-intrinsic base weight and action

$$m_r : Q_r \rightarrow \mathbb{R}_{>0}, \quad S_r : Q_r \rightarrow \mathbb{R} \cup \{+\infty\},$$

and

$$w_r(q) = m_r(q) e^{-S_r(q)}, \quad Z_r = \sum_{q \in Q_r} w_r(q), \quad \mu_r(q) = Z_r^{-1} w_r(q).$$

Equivalently, one may state an intrinsic projective prior  $\nu_r$  on  $Q_r$  and set  $\mu_r = (n_r)_{\#} \nu_r$ . Uniform quotient counting, uniform representative counting pushed to the quotient, groupoid weights, and tracial central-sector weights are different physical claims. The paper must declare which one is being used.

**Normal-form projector non-selection.** For any retraction  $N : Q \rightarrow Q_{\text{nf}}$  onto a subset  $Q_{\text{nf}} \subseteq Q$ , that is, any map whose restriction to  $Q_{\text{nf}}$  is the identity, the induced map on laws

$$\mathcal{C}_Q(\mu) = N_{\#} \mu$$

is idempotent:

$$\mathcal{C}_Q^2 = \mathcal{C}_Q.$$

Every law supported on  $Q_{\text{nf}}$  is fixed; both statements use the retraction property. Therefore settlement or canonicalization never selects a unique physical probability law by itself.

**Selection-gap corollary.** Let  $X \subseteq Q_{\text{nf}}$  be a finite set of quotient-normal candidates distinguished by visible invariants. Normal-form data determine  $X$  and its quotient-visible invariants, but they do not choose a member of  $X$ . If two laws  $\mu, \nu$  are supported on  $X$  and concentrate on different candidates, both are fixed by  $\mathcal{C}_Q$ . Unique sector selection therefore requires source data: an intrinsic action with a unique minimizer, a declared physical ensemble, or a refinement-stable gap certificate. A defect or holonomy classification can classify possible sectors without choosing the physical sector, and a contraction or repair generator can certify convergence toward a declared target without creating the target law.

**Finite MaxEnt quotient ensemble.** For finite  $Q$ , positive  $m$ , and quotient observables  $F_1, \dots, F_k$ , maximizing

$$\mathcal{H}_m(\nu) = - \sum_q \nu(q) \log \frac{\nu(q)}{m(q)}$$

subject to

$$\sum_q \nu(q) = 1, \quad \sum_q \nu(q) F_a(q) = c_a$$

has the full-support solution, when the feasible full-support surface is nonempty,

$$\mu(q) = \frac{m(q) \exp[-\sum_a \theta_a F_a(q)]}{Z(\theta)}.$$

Boundary optima obey the same formula after restricting to their support. On a finite noncommutative quotient algebra with faithful reference state  $\sigma_r$ ,

$$\rho_r = \frac{\exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}{\text{Tr} \exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}.$$

The finite constraint ledger must name every  $F_{r,a}$ , its units and support, the target expectation and source, sector or zero-mode treatment, refinement transformation, and proof that no run output or observational output entered the source definition.

**Refinement compatibility and RG closure.** For  $s \succeq r$ , let  $c_{sr} : Q_s \rightarrow Q_r$  be the physical coarse map. Exact compatibility of weighted ensembles is equivalent to the fiber-sum identity

$$\sum_{q': c_{sr}(q')=q} m_s(q') e^{-S_s(q')} = \alpha_{sr} m_r(q) e^{-S_r(q)}$$

for a constant  $\alpha_{sr} > 0$  independent of  $q$ . Then

$$(c_{sr})_{\#} \mu_s = \mu_r.$$

If the one-step defects are

$$\delta_{k+1,k} = \|(c_{k+1,k})_{\#} \mu_{k+1} - \mu_k\|_{\text{TV}},$$

then

$$\|(c_{nr})_{\#} \mu_n - \mu_r\|_{\text{TV}} \leq \sum_{k=r}^{n-1} \delta_{k+1,k}.$$

For exponential-family refinement, exact closure requires the fine conditional free energy

$$G_{sr,\theta}(q_r) = -\log \mathbb{E}_{m_s^0} [\exp[-\theta \cdot F_s(Q_s)] \mid c_{sr}(Q_s) = q_r]$$

to equal  $\kappa_{sr}(\theta) + R_{sr}(\theta) \cdot F_r(q_r)$ . If the residual is uniformly bounded by  $\varepsilon$ , the induced total-variation defect is bounded by  $\tanh \varepsilon$ .

**Implementation invariance and representative lifting.** If implementations  $A, B$  have quotient bijections  $h_r : Q_r^A \rightarrow Q_r^B$  satisfying

$$m_r^B(h_r q) = m_r^A(q), \quad S_r^B(h_r q) = S_r^A(q), \quad h_r \circ c_{sr}^A = c_{sr}^B \circ h_s,$$

then

$$(h_r)_\# \mu_r^A = \mu_r^B.$$

For tracially pointed quantum quotients the corresponding equivalence is a trace-preserving quotient equivalence. It is invariant under unitary intertwiners preserving the gauge action and sector, and under inert trivial ancillas  $A \mapsto A \otimes I_{\text{anc}}$ . It is not invariant under arbitrary changes of gauge-representation multiplicities.

If an implementation stores representatives, a representative-level law must be a conditional lift

$$\tilde{\mu}_r(x) = \mu_r(\pi_r x) \kappa_r(x \mid \pi_r x), \quad \sum_{x: \pi_r(x)=q} \kappa_r(x \mid q) = 1.$$

Then  $(\pi_r)_\# \tilde{\mu}_r = \mu_r$ . Uniform representative sampling yields orbit-size weights and is physical only if representative counting is the declared base measure.

**Quotient-lumpable kernels and sampler correctness.** A representative kernel  $\tilde{P}(x, y)$  descends to  $Q_r$  only when

$$P_Q(q, q') = \sum_{y: \pi(y)=q'} \tilde{P}(x, y)$$

is independent of the chosen representative  $x \in \pi^{-1}(q)$ . For  $w(q) = m(q)e^{-S(q)}$  and proposal  $R(q, q')$  with reciprocal support, the Metropolis–Hastings acceptance rule

$$a(q, q') = \min \left\{ 1, \frac{w(q')R(q', q)}{w(q)R(q, q')} \right\}$$

gives detailed balance

$$\mu(q)R(q, q')a(q, q') = \mu(q')R(q', q)a(q', q).$$

Repair-informed proposals must include the Hastings asymmetry term; otherwise the stationary law is generically changed.

**Repair generators are not selectors.** A repair generator of the form

$$L_{\text{rep}} = \sum_C c_C (I - E_C)$$

is a relaxation or sampling object after a law has been selected. Conditional expectations  $E_C$  are defined on  $L^2(X_r, \pi_r)$ , so the reference law  $\pi_r$  is input. On overlapping collars the expectations need not commute. The correct finite gap certificate is the Poincare constant

$$\kappa_r = \inf_{f \perp 1} \frac{\sum_C \|(I - E_C)f\|^2}{\|f\|^2}.$$

If local fiber rates have a positive lower bound  $\gamma_*$ , then

$$L_{\text{rep}} \geq \gamma_* \kappa_r (I - P_0).$$

Finite repair completeness gives  $\kappa_r > 0$  at fixed regulator. A uniform refinement lower bound  $\inf_r \kappa_r > 0$  is a separate theorem or receipt.

**Finite evidence accuracy.** For bounded coarse observables  $O$ , if

$$\|\hat{\mu}_s - \mu_s\|_{\text{TV}} \leq \epsilon_{\text{samp}}$$

and the refinement defects sum to  $\epsilon_{\text{ref}}$ , then

$$\left| \mathbb{E}_{\hat{\mu}_s}[O \circ c_{sr}] - \mathbb{E}_{\mu_r}[O] \right| \leq 2\|O\|_{\infty}(\epsilon_{\text{samp}} + \epsilon_{\text{ref}}).$$

Continuum-facing observables require a realization map and correlation Cauchy bound in addition to a finite histogram.

**Vacuum promotion gate.** A stationary sampler is not a physical vacuum. For any faithful target law one can build a positive transfer operator with that law as ground state, so positivity alone is not a selector. Vacuum promotion requires source Euclidean slab data

$$\mathfrak{S}_r^E = (Q_r, m_r^0, J_r, V_r, a_{t,r})$$

whose conductance  $J_r(q, q') = J_r(q', q) \geq 0$ , local potential  $V_r$ , and slab thickness  $a_{t,r}$  are derived without using the target law or sampler output. With connected event graph,

$$(H_r^E f)(q) = \frac{1}{m_r^0(q)} \sum_{q'} J_r(q, q') (f(q) - f(q')) + V_r(q) f(q)$$

is self-adjoint and bounded below on  $L^2(Q_r, m_r^0)$ ; its finite Feynman–Kac semigroup is positivity improving. Perron–Frobenius gives a unique positive normalized ground state  $\Omega_r$ , and the finite vacuum law is

$$\mu_r^{\text{vac}}(q) = |\Omega_r(q)|^2 m_r^0(q).$$

For  $T_r = e^{-a_{t,r}(H_r^E - E_{0,r})}$ , the Doob kernel is stochastic and detailed-balanced with  $\mu_r^{\text{vac}}$ . Continuum promotion additionally requires reflection positivity or equivalent reconstruction plus refinement compatibility of the transfer family.

**Primordial and cosmological prediction firewall.** A screen covariance contains incomplete radial information. The complete one-shell map

$$C_\ell = 4\pi \int_0^\infty \frac{dk}{k} \Delta_\zeta^2(k) j_\ell^2(k\chi_\star)$$

has an infinite-dimensional kernel that persists under positivity. OPH primordial promotion requires the source-only stress, single-clock, entropy-repair, curvature-evolution, adiabatic-mode, phase-coherence, physical mode, radial-null-space, and forward-projection receipts together with a scale-natural physical dilation intertwiner or complete radial cross-covariance tomography. A finite radial prior produces a conditional continuation. Observable CMB comparison also requires declared source, solver, dataset, covariance, nuisance, data-use, and pooled-reducer provenance.

**Claim tiers and required receipts.** Every ensemble-facing run records its ensemble id, claim tier, regulator, representative schema, gauge action, canonicalizer, base measure, action coefficients, coarse maps, zero-mode projector, amplitude convention, sampler, smoothing policy, source

provenance, and explicit nonclaims. The seed belongs to the run receipt rather than the ensemble definition. The claim tiers are

- $E0$  : seed noise, proposal noise, repair jitter,
- $E1$  : conventional reference ensemble,
- $E2$  : OPH-native quotient ensemble,
- $E3$  : OPH vacuum,
- $E4$  : OPH primordial field,
- $E5$  : observable cosmological prediction.

The evidence bundle must keep separate receipts for stationary-law schedule invariance, detailed balance of the aggregate kernel, and pathwise partition invariance. Deterministic replay of semantic random streams or a canonical serial chain is useful, but it is not pathwise partition invariance. Smoothing must preserve raw coefficients, raw spectra, smoothing kernels, smoothed coefficients, smoothed spectra, and hashes of each stage; it is not part of  $S_r$  unless explicitly declared.

### 3 Vacuum Claim Boundary

The strongest allowed simulator labels are:

- $E0$  : seed noise or repair jitter,
- $E1$  : conventional reference ensemble.

An OPH-native vacuum label requires an  $E3$  receipt. The source data must include

$$\mathfrak{S}_r^E = (Q_r, m_r^0, J_r, V_r, a_{t,r})$$

with  $J_r$ ,  $V_r$ , and  $a_{t,r}$  derived from OPH source structure rather than from a target law or a desired visual spectrum. Accepted repair moves are generally directed and cannot simply be reused as reversible Euclidean conductances.

### 4 Structure-Formation Requirements

| Object                   | Required source-side label                                                                                                                                |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fluctuating screen field | $E0$ or $E1$ unless an $E2$ quotient ensemble receipt passes.                                                                                             |
| OPH vacuum field         | $E3$ only after source Euclidean slab, ground-state, Doob, reflection-positivity, and transfer-refinement receipts.                                       |
| Proto-object worldlines  | Diagnostic until production matter/particle and observer-facing object population receipts pass.                                                          |
| Growth seeds             | $E4$ only after the primordial bridge, physical mode basis, radial-null, positivity, forward-projection, and source-dilation or tomography receipts pass. |
| LSS comparison           | Diagnostic; source, transfer, and official likelihood/data contracts are work in progress.                                                                |

## 5 Simulator Contract

The simulator should emit the following receipt names even when they fail:

- TRACIALLY\_POINTED\_QUOTIENT\_RECEIPT
- SOURCE\_EUCLIDEAN\_SLAB\_RECEIPT
- TRANSFER\_REFINEMENT\_RECEIPT
- STATIONARY\_LAW\_SCHEDULE\_INVARIANCE\_RECEIPT
- PATHWISE\_PARTITION\_INVARIANCE\_RECEIPT

False receipts are part of the scientific result. They prevent a reference baseline or rendering field from being mislabeled as an OPH vacuum.

## 6 Unclosed Claim Boundaries

1. Derive the reversible source conductances  $J_r$ , potential  $V_r$ , and slab thickness  $a_{t,r}$  from finite OPH source data.
2. Prove reflection positivity or an equivalent reconstruction theorem on the selected branch.
3. Relate passed E3 vacuum receipts to E4 primordial growth seeds without using CMB or LSS data.
4. Specify proto-object/worldline receipts that remain distinct from production particle claims.